

The Embui Protocol: Designing a Merit-Based Universal Basic Income to Complement Artificial General Intelligence for Human Advancement

by Stephen Makonin



©SHUTTERSTOCK.COM/ARTEMISDIANA

It is apparent that artificial general intelligence (AGI) and large language models (LLMs) are and will increasingly displace traditional labour. Some might urgently advocate the need for a universal basic income (UBI). Rather than viewing UBI as a static entitlement, this article proposes a dynamic, merit-based system that rewards individuals for their meaningful contributions to societal progress and AGI development. Drawing on metadata attribution, Shareability Index (S-index)-inspired metrics, and scalable activity-based tiers, The Embui Protocol outlines a policy framework to ensure equitable compensation, dignity through

participation, and alignment of human effort with the advancement of AGI toward humanwide betterness.

Introduction: Rethinking UBI in the Age of AGI

We are entering an era where machines can write, reason, and respond in ways once limited to human intelligence. AGI and its practical cousin—LLMs—are reshaping industries, redefining productivity, and raising existential questions about the role of human labour. Automation is no longer about replacing muscle; it is replacing cognition, creativity, and coordination.

Discussions about UBI have gained popularity as a defence against these disruptions, offering a government safety net to cushion the socioeconomic fallout. However, most UBI models are set amounts and passive recurring payments (i.e., social system entitlements). They focus on subsistence rather than empowerment and risk reinforcing dependency on a government social service. We believe that UBI needs to promote contribution to reinforce self-worth and the feeling of accomplishment.

This article proposes an alternative: a merit-based UBI framework that actively rewards individuals for contributing to the very systems that are reshaping our world. By leveraging AGI systems to evaluate and attribute human inputs—such as volunteerism, prompt engineering,

and think-tank solutions—we can create a feedback loop that not only compensates but inspires. This approach, *The Embui Protocol*, reimagines UBI not as a static entitlement, but as an engine of participatory advancement.

Embui Etymology

Embui (pronounced *uhm-byu*) is the acronym for *merit-based universal income*. A take on the word *imbue* (i.e., swapping of the “i” and the “e”), a verb meaning to fill something, typically a person or object, with a particular feeling, quality, or idea. In this case, human effort *imbues* AGI betterment.

The Case for Merit-Based UBI

UBI, rooted in merit, recognizes the intrinsic value of being good or worthy—

not in a moralistic sense, but in measurable, socially impactful ways. Unlike current models that treat all recipients equally regardless of activity, a merit-based system acknowledges effort, expertise, and contribution (Figure 1).

Analogous systems already exist. Open source developers gain reputation through GitHub contributions.

UBI, rooted in merit, recognizes the intrinsic value of being good or worthy—not in a moralistic sense, but in measurable, socially impactful ways.

Stack Overflow users earn reputation points through helpful answers. Wikipedia editors are acknowledged for curating knowledge. These platforms have proven that recognition can drive meaningful participation—even without monetary reward.

Evidence from UBI experiments further strengthens this view. In Finland’s UBI pilot, participants reported higher levels of happiness and trust without reducing their work participation [1]. Likewise, experiments in Kenya and India show improvements in well-being and social participation when people receive unconditional income [2]. A study by Hidalgo et al. [3] found that neither UBI nor taxation on robots reduced workers’ motivation to exert effort—indicating that basic income can coexist with ongoing productive engagement.

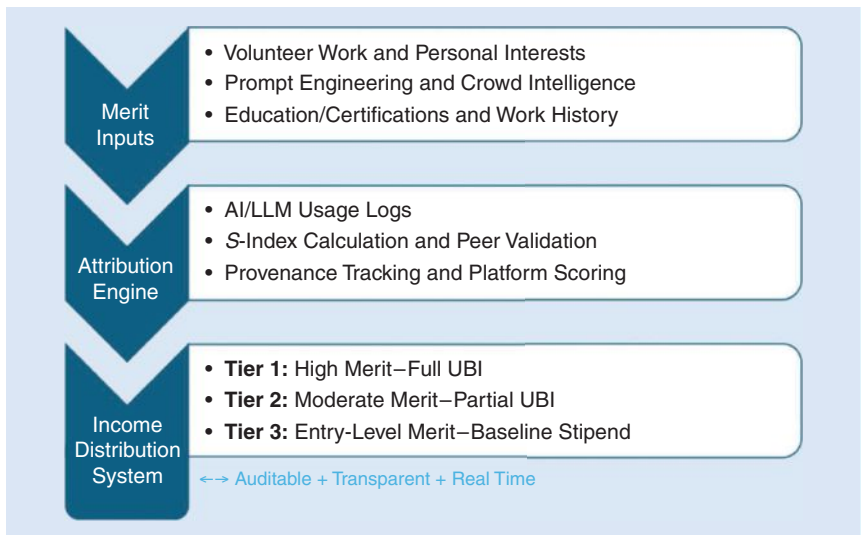


Figure 1: The Embui Protocol: inputs, attribution, and income flow. AI: artificial intelligence.

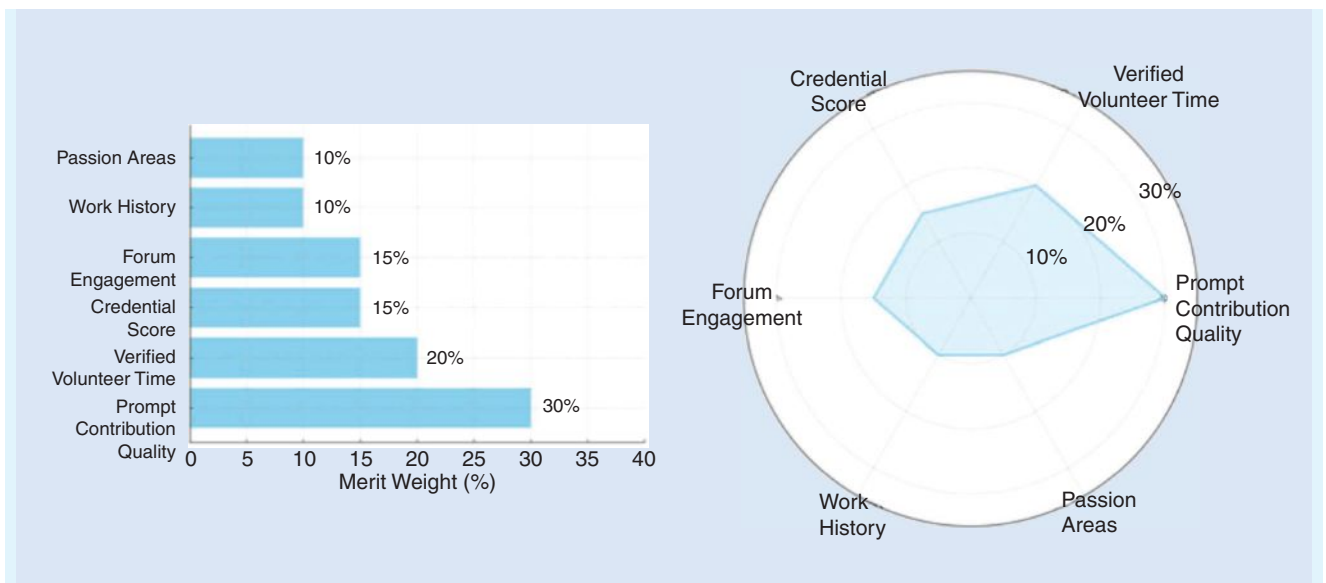


Figure 2: A sample merit portfolio composition for The Embui Protocol.

With AGI and LLMs increasingly trained on human-created data and prompts, there is a clear opportunity: those who help build, refine, or contextualize these models should be rewarded for their labour and insight. This aligns economic reward with intellectual contribution and democratizes participation in the future of intelligence.

Merit-based UBI ensures dignity. It provides a pathway for individuals to continue contributing—even after traditional employment becomes scarce—by valuing their ideas, guidance, and domain knowledge in shaping artificial intelligence (AI) systems.

Defining “Merit” in the AGI Ecosystem

Merit is a measure of being good or worthy, which can increase over time based on activities that align with societal progress and responsible AGI development. For an AGI to be equitable, merit must be quantifiable, transparent, and inclusive. Three key merit channels emerge:

1. *Volunteerism*: This is contributions to education, social services, environmental sustainability, and other altruistic activities that enhance communal well-being. According to the United Nations [4], volunteer work globally contributes the equivalent of 2.4% of world gross domestic product, demonstrating enormous—often unpaid—societal value.
2. *Prompt engineering*: This is writing, refining, and testing prompts that improve LLM performance, safety, and interpretability. Federiakin et al. [5] define prompt engineering as a critical 21st century skill, highlighting its growing role in both personal and professional contexts.
3. *Crowd intelligence*: This is participation in collaborative problem-solving platforms, such as open think tanks, scientific prediction markets, or idea incubators.

Merit must be tracked via metadata. Just as Git commits or academic citations trace intellectual lineage, AGI systems must include attribution layers—tracking who influenced what and how. A modified version of the S-index [6] could measure how widely and usefully a person’s contributions are used in AI outputs, policy recommendations, or community tools.

Economic literature has also advocated treating user-generated data and intellectual contributions as labour. Arrieta-Ibarra et al. [7] argues that users who contribute to digital platforms should be compensated fairly for the data they

provide, laying the groundwork for compensating prompt engineering and other digital labour.

Income Tiers and Scaling Mechanism

A key distinction of The Embui Protocol is that UBI is not one size fits all. It scales based on a combination of dynamic and historical factors: *current participation* functionality that can track verified,

By compensating humans for their insight, creativity, and care, we steer AGI toward amplifying the best of humanity, rather than making these types of productivity obsolete.

high-quality engagement in platforms that support AGI, social good, or open data; *past achievements* awareness of educational credentials, certifications, work history, and open source or scientific contributions; *volunteer record* documented service of pro-social behaviour; and *personal interests* settings for alignment with long-term learning goals, domain-specific expertise, or creative pursuits.

In 2019, Guzman et al. [8] published the CarbonKit framework, which introduced an *eco-currency* (i.e., tradable alternative carbon currency) that users earn by reducing their personal carbon footprint through actions like opting for public transit, lowering

home energy use, and meeting step-count goals, with balances redeemable for low-emission products, tax rebates, or in-kind incentives.

Each individual’s “merit portfolio” (Figure 2) would be updated in real time and evaluated by transparent AI-driven systems trained to identify authentic, impactful efforts. Peer-review mechanisms, public dashboards, and open source auditing tools would ensure accountability and fairness.

This tiered model (Figure 1) respects the diversity of contribution: not everyone has the same resources, but everyone can participate meaningfully.

Infrastructure Blueprint: What Needs to Exist

Implementing a merit-based UBI requires a robust technological and institutional foundation having the following features:

- *Digital identity frameworks* to verify credentials (e.g., blockchain-based or IEEE-issued digital badges) to ensure authenticity.
- *Metadata-aware LLMs* with next-generation models that retain prompt lineage and attribute influence with granularity.
- *Contribution platforms* that use secure, decentralized hubs where users can submit prompts, solutions, or volunteer logs.
- *Evaluation engines* for AI systems that assess quality, originality, and societal impact based on open metrics that are auditable with trackable history.

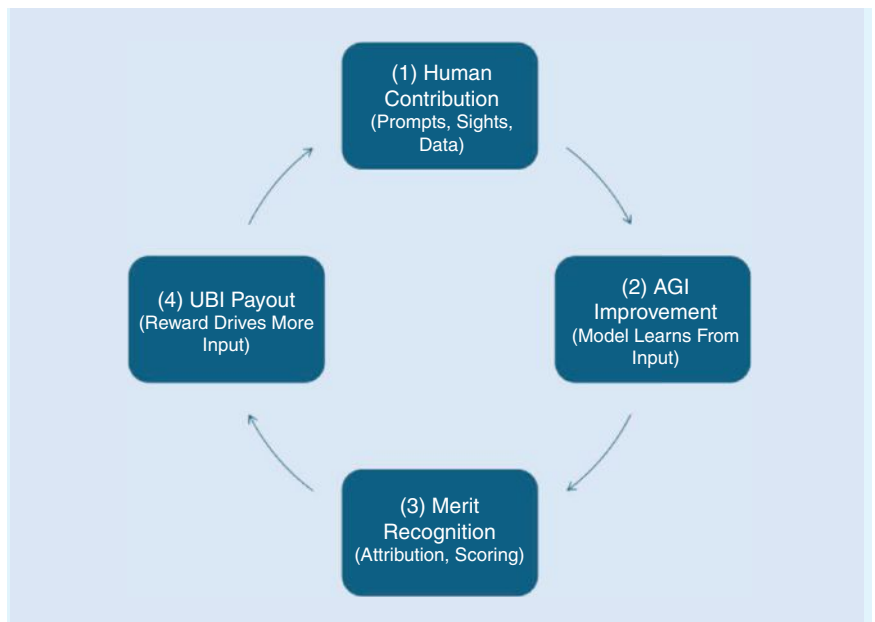


Figure 3: AGI–human alignment via The Embui Protocol.

- *Payment layers* that provide crypto-agnostic or fiat-compatible smart contracts for distributing micro-UBI disbursements based on accumulated merit.

These components must be interoperable, inclusive, and governed by an independent standards body—potentially under IEEE’s Technology and Society umbrella—to ensure ethical oversight and global applicability.

Societal Outcomes and AGI Alignment

A meritocratic UBI framework does more than redistribute wealth, it realigns incentive structures toward human flourishing (Figure 3). It encourages

- *Lifelong learning*: Citizens are rewarded for engaging in ongoing skill development and knowledge contribution.
- *Collaborative intelligence*: Think-tank models replace siloed competition with transparent, collective problem-solving.
- *Ethical AGI shaping*: AI systems evolve based on well-attributed, community-sourced knowledge and values.
- *Inclusion*: Historically marginalized communities gain access to recognition and income by participating in nontraditional ways.

Research by Eloundou et al. [9] estimates that up to 80% of jobs may be affected by LLMs like Generative Pre-trained Transformer 4, with nearly one-fifth potentially seeing more than 50% task replacement. Such findings underscore the urgent need for alternative compensation models tied to intellectual contribution and insight.

The World Bank’s 2019 World Development Report has already recommended decoupling social support from formal employment in light of these shifts—further legitimizing merit-based UBI as a forward-thinking solution [10].

By compensating humans for their insight, creativity, and care, we steer AGI toward amplifying the best of humanity, rather than making these types of productivity obsolete.

Policy Recommendations and Path Forward

The IEEE Technology and Society community has a pivotal role in building out and developing this new paradigm [11]. The recommended next steps include the following five key tasks:

1. Develop a prototype merit-tracking framework using IEEE digital badges and metadata standards.
2. Launch a call for contributors to pilot merit attribution in prompt engineering, open source AI, and volunteer platforms.
3. Propose policy standards for AGI developers to integrate attribution-aware architectures.
4. Create open datasets that track impact and influence of prompts and ideas within LLM training cycles.
5. Engage with governments and nongovernmental agencies to co-develop hybrid funding models for scaled UBI deployment.

The transition to a post-labour economy is inevitable [12]. The question is not *if*, but *how* we will support the dignity, creativity, and insight of those displaced. The Embui Protocol offers a way forward: a society where income follows merit, merit fuels AGI, and AGI propels humanity forward. ■

References

- [1] C. Arnold, “Money for nothing: The truth about universal basic income,” *Nature*, vol. 557, no. 7707, pp. 626–628, 2018, doi: 10.1038/d41586-018-05259-x.
- [2] A. Banerjee, P. Niehaus, and T. Suri, “Universal basic income in the developing world,” *Annu. Rev. Econ.*, vol. 11, pp. 959–983, Aug. 2019, doi: 10.1146/annurev-economics-080218-030229.

- [3] A. Cabrales, P. Hernández, and A. Sánchez, “Robots, labour markets, and universal basic income,” *Humanities Soc. Sci. Commun.*, vol. 7, no. 9, 2020, Art. no. 185, doi: 10.1057/s41599-020-00676-8.
- [4] United Nations Volunteers, “State of the world’s volunteerism report 2018,” UNV, Bonn, Germany, 2018. [Online]. Available: <https://www.unv.org/publications/swvr2018>
- [5] D. Federiakin et al., “Prompt engineering as a new 21st century skill,” *Front. Educ.*, vol. 9, Nov. 2024, Art. no. 1366434, doi: 10.3389/educ.2024.1366434.
- [6] S. Makonin, “Editorial: Hey Google Scholar, let’s calculate the Sharability Index!” *IEEE Data Descriptions*, vol. 2, p. 26, 2025, doi: 10.1109/IEEEDATA.2025.3551932.
- [7] I. Arrieta-Ibarra, L. Goff, D. Jiménez-Hernández, V. Lanier, and E. G. Weyl, “Should we treat data as labour? Moving beyond ‘free,’” *AEA Papers Proc.*, vol. 108, pp. 38–42, May 2018, doi: 10.1257/pandp.20181003.
- [8] L. Guzman, S. Makonin, and A. Clapp, “CarbonKit: Designing a personal carbon tracking platform,” in *Proc. 4th Int. Workshop Social Sens.*, 2019, pp. 24–29, doi: 10.1145/3313294.3313385.
- [9] T. Eloundou et al., “GPTs are GPTs: An early look at the labour market impact potential of large language models,” 2023, *arXiv:2303.10130*.
- [10] “World development report 2019: The changing nature of work,” World Bank, Washington, DC, USA, 2019. [Online]. <https://www.worldbank.org/en/publication/wdr2019>
- [11] M. Polavarapu, “The future AI isn’t what it used to be: AI’s rapid evolution and SSIT’s role in shaping tomorrow,” *IEEE Technol. Soc. Mag.*, vol. 44, no. 1, p. 4, Mar. 2025, doi: 10.1109/MTS.2025.3534607.
- [12] Y. Shen and X. Zhang, “The impact of artificial intelligence on employment: The role of virtual agglomeration,” *Humanities Soc. Sci. Commun.*, vol. 11, Jan. 2024, Art. no. 122, doi: 10.1057/s41599-024-02647-9.

About the Author

Stephen Makonin (smakonin@sfu.ca) received his Ph.D. degree in computing science from Simon Fraser University (SFU), Burnaby, Canada, in 2014. He is an adjunct professor at the School of Engineering Science and the principal investigator of the Computational Sustainability Lab, SFU, Vancouver, Canada. He holds various volunteer positions at IEEE (e.g., IEEE DataPort and IEEE SA P2957 BDGMM). He is a registered professional engineer with EGBC and has more than 30 years of software and data engineering experience in both industry and academia. In 2024, he became the founding editor-in-chief of *IEEE Data Descriptions*. He is a Senior Member of IEEE.